Wildfire WUI Risk Model — Technical Documentation

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Overview

The project builds a structure-level wildfire risk model for the wildland-urban interface (WUI) from fuels, topography, fire history and building footprints, then tests it against observed structure losses in a recent fire that the model never saw.

Research questions

1. How well do fuel, terrain, fire-history and built-environment features predict whether a structure exposed to fire is destroyed?
2. How much does performance drop under honest spatial cross-validation compared with random cross-validation?
3. Would the model have ranked the structures actually lost in a held-out recent fire as high risk?
4. Which variables drive predictions, and how do they interact (e.g. slope × fuel, building density × distance to wildland)?

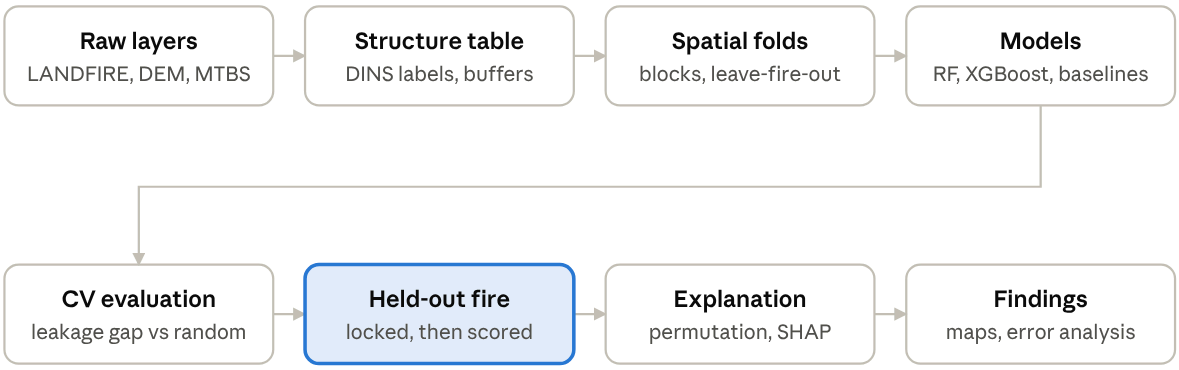
Contribution — (a) validation against real, post-event structure damage data rather than burn probability alone; (b) a spatial-CV protocol that quantifies autocorrelation leakage; (c) an explanation of variable importance with SHAP and partial dependence, checked for stability across folds.

Scope — California WUI as the study area (adjustable). Two prediction targets are kept distinct: *exposure* (does fire reach the parcel) and *conditional loss* (given exposure, is the structure destroyed). The primary model is conditional loss, because structure damage inspection data labels exposed structures. Out of scope: fire spread simulation, ember transport physics and real-time weather forecasting.

Pipeline

The analysis runs in eight stages; the held-out fire is excluded from every stage until predictions for it are locked.

The test fire stays unseen until its predictions are locked



analysis pipeline · 8 stages

The spatial folds feed both hyperparameter tuning and evaluation, and the same folds are reused by every model so comparisons are paired.

Data sources and preprocessing

All layers are reprojected to one equal-area CRS (California Albers, EPSG:3310) and features are taken at the version available *before* each fire, so no post-fire information leaks into predictors.

Layer	Source	Resolution	Use
Fuel models, canopy cover, canopy height, canopy base height, existing vegetation type	LANDFIRE	30 m raster	Fuel features in buffers around each structure
Elevation, slope, aspect	USGS 3DEP DEM	10–30 m	Terrain features, topographic position
Fire perimeters (large fires)	MTBS	Polygons + burn severity	Fire history, exposure definition
Fire perimeters (all CA)	CAL FIRE FRAP	Polygons	Fire history, recent-fire perimeter
Structure damage labels	CAL FIRE DINS (Damage Inspection)	Points	Target: destroyed vs not destroyed

Layer	Source	Resolution	Use
Building footprints	Microsoft / Overture building footprints	Polygons	Structure geometry, density, spacing
Roads	OpenStreetMap / TIGER	Lines	Access, distance to road
WUI class	SILVIS Lab WUI maps	Census blocks	Stratification, interface vs intermix
Fire weather (optional)	gridMET	~4 km daily	Event wind, VPD, ERC for conditioning

Source versions and download dates are pinned in `data/manifest.yaml`; verify current release names and licences before use.

Preprocessing steps

1. **Version matching** — for each fire, use the LANDFIRE release whose reference year precedes ignition (e.g. LF 2020 or 2022 for a 2025 fire), and footprints captured before the event.
2. **Label join** — match DINS points to footprints (nearest footprint within 30 m; unmatched points kept as point geometry). Collapse damage classes to binary: *Destroyed (> 50%)* = 1; *No damage, Affected, Minor, Major* = 0. A sensitivity run uses *Major + Destroyed* = 1.
3. **Exposure set** — only structures inside the final fire perimeter, or with a DINS record, enter the conditional-loss dataset.
4. **Cleaning** — drop duplicate DINS records, non-residential outbuildings < 20 m² (sensitivity check), and footprints with invalid geometry.
5. **Raster sampling** — zonal statistics in buffers (see features), computed with `exactextract` for partial-pixel weighting.

Unit of analysis, labels and features

The unit is one structure (building footprint); every feature describes that structure's surroundings at multiple radii, because structure loss depends on both the immediate home ignition zone and the wider landscape.

Label — `destroyed` $\in \{0, 1\}$ from DINS, for exposed structures. Expect class imbalance that varies widely between fires; report base rates per fire.

Buffers — 30 m (home ignition zone), 100 m, 500 m, and 2 km rings around each footprint.

Group	Features	Notes
Fuels	% of buffer in each fuel group (grass, grass-shrub, shrub, timber understory, timber litter, slash, non-burnable); mean canopy cover, canopy height, canopy base height	From LANDFIRE FBFM40 and canopy layers, per buffer
Fuel continuity	Distance to nearest contiguous wildland patch > 1 ha; edge density of burnable fuel	Captures interface vs intermix
Terrain	Slope, aspect (as northness and eastness), elevation, topographic position index (TPI) at 300 m and 2 km, ruggedness	Mid-slope and ridge positions behave differently
Alignment	Upslope/downslope position relative to wildland; slope × fuel interaction	Optional event-specific wind alignment when weather is included
Built environment	Structures within 30/100/500 m, distance to nearest structure, footprint area, building separation	Structure-to-structure spread is a major loss driver in dense areas
Access	Distance to nearest road, road density, distance to fire station	Proxy for suppression access
Fire history	Years since last burn within 500 m; number of fires since 1984; prior burn severity	From MTBS + FRAP
Event weather (optional)	Max wind speed, VPD, ERC on ignition day(s)	Only in the event-conditioned model variant
Building attributes (if available)	Year built, roof/vent class from DINS or assessor	Year built proxies building code era; often missing, so kept in a separate model variant

Leakage rules — no feature may use the fire's own perimeter, severity or post-fire imagery; distance-to-perimeter features are excluded; footprint density is computed from pre-fire footprints only.

Correlation screening — features with |Spearman ρ| > 0.9 are clustered and one representative is kept, so importance is not split arbitrarily between near-duplicates.

Models

Random forest and XGBoost are trained on the same features and folds and compared against simple baselines, so any gain is attributable to the learner rather than the setup.

Baselines

- Logistic regression on standardized features (with slope × fuel interaction).
- Single-variable rules: structure density only; distance to wildland only.
- An existing hazard product (e.g. CAL FIRE Fire Hazard Severity Zone class) as an ordinal predictor.

Random forest — scikit-learn `RandomForestClassifier`, 1 000 trees, `class_weight="balanced_subsample"`. Tuned: `max_features` {sqrt, 0.3, 0.5}, `min_samples_leaf` {1, 5, 20}, `max_depth` {None, 20}.

XGBoost — `XGBClassifier`, `tree_method="hist"`, early stopping on an inner spatial fold. Tuned: `max_depth` 3–8, `learning_rate` 0.02–0.1, `subsample` 0.6–1.0, `colsample_bytree` 0.5–1.0, `min_child_weight` 1–20, `scale_pos_weight` = negatives/positives in the training fold. Optional monotone constraints (e.g. risk non-decreasing with slope) as a sensitivity run.

Tuning — nested spatial CV: outer folds estimate performance; inner spatial folds (inside each outer training set) choose hyperparameters with Optuna, 100 trials, optimizing PR-AUC.

Calibration — isotonic regression fit on inner-fold out-of-fold predictions, so output probabilities are interpretable as loss likelihood given exposure.

Metrics

Metric	Why
ROC-AUC	Ranking quality, comparable to prior studies

Metric	Why
PR-AUC	Ranking quality under class imbalance (primary)
Brier score + reliability curve	Calibration
Recall at top 10% / 20% of ranked structures	Practical value for targeting mitigation
Log loss	Probabilistic accuracy

Uncertainty: 95% intervals by spatial block bootstrap (resampling blocks, not structures).

Spatial cross-validation

Neighbouring structures share fuels, terrain, weather and often the same fate, so random CV lets the model memorise neighbourhoods; every reported number therefore comes from folds that separate training and test data in space, and in fire event.

Step 1 — measure autocorrelation. Fit an empirical variogram (or Moran's I correlogram) of the residuals of a simple model and of key features. The range at which autocorrelation drops below 0.1 sets the minimum block size and buffer distance r.

Step 2 — fold schemes compared

Scheme	How	What it tests
Random k-fold (reference only)	Structures shuffled, k = 5	Optimistic upper bound; shows leakage
Spatial block CV	Square blocks of side ≥ r, assigned to 5 folds	Generalisation to new neighbourhoods
Buffered leave-one-block-out	Test block plus a buffer of width r removed from training	Removes edge leakage between adjacent blocks
Leave-one-fire-out (LOFO)	All structures of one fire held out together	Generalisation to a new fire event (closest to real use)

LOFO is the primary scheme; spatial block CV is reported alongside it. The **leakage gap** = random-CV metric − LOFO metric is reported as a headline result.

Rules

- Hyperparameter tuning, feature screening, calibration and class weighting happen inside each training fold only.
- Fires with fewer than 30 destroyed structures are grouped with fires from the same region/year for LOFO to keep test folds informative.
- Folds are saved (`folds.parquet` , structure ID → fold) so all models use identical splits.

Implementation: `spatialkfold` / custom GeoPandas block assignment, scikit-learn `GroupKFold` with block or fire IDs as groups.

Held-out test: a recent fire

The central test freezes the final model on fires *before* a cutoff date, then scores every exposed structure in one recent fire and compares the ranking with the inspected losses.

Choosing the test fire — a recent California WUI fire with a published DINS dataset, several hundred or more inspected structures, and a mix of destroyed and surviving homes. Candidates include the January 2025 Eaton and Palisades fires in Los Angeles County; confirm data availability and record counts before committing. Running both allows a check on whether results transfer between two very different settings (dense urban-edge vs. coastal-canyon WUI).

Protocol

1. Training set = all fires that ignited before the test fire's ignition date; no data from the test fire or later is used for features, tuning or calibration.
2. Features for test structures use pre-fire LANDFIRE and footprints only.
3. Score all structures inside the perimeter (and DINS records outside it).
4. Lock predictions (hash and timestamp the file) before opening the labels.

Evaluation

- ROC-AUC, PR-AUC, Brier score with block-bootstrap intervals.
- Lift curve: share of destroyed structures captured in the top 10%, 20% and 50% of predicted risk.
- Comparison with baselines and with the official hazard zone class.
- Maps: predicted risk, observed loss, and residual (false negatives and false positives) clusters.

- Stratified results: by WUI class (interface vs intermix), building density tercile and distance to wildland.

Error analysis — false-negative clusters are examined against event conditions the model does not see (extreme wind, ember-driven structure-to-structure spread, suppression resource saturation). This separates *model* failures from *predictor-set* limits, and is the main interpretive contribution of the test.

Variable importance and explanation

Importance is reported only from held-out folds and only where it is stable across folds, because tree-based importance is easily distorted by correlated and high-cardinality features.

Methods

Method	Computed on	Answers
Permutation importance (drop in PR-AUC)	Each outer test fold	How much performance depends on a feature
Grouped permutation importance	Feature groups (fuels, terrain, built environment, access, history)	Which *kind* of information matters, robust to within-group correlation
SHAP (TreeSHAP)	XGBoost, test folds	Direction and size of each feature's effect per structure
SHAP interaction values	Top 10 features	Key interactions, e.g. slope × shrub fraction
Partial dependence / ALE plots	Top features	Shape of the response; ALE preferred where features correlate

Impurity-based (Gini/gain) importance is reported only as a supplementary table, with a note on its bias.

Stability — rank each feature in every outer fold; report median rank and interquartile range. A feature is called important only if it is in the top 10 in at least 80% of folds.

Expected questions the analysis should answer

- Does structure density/spacing outrank wildland fuel measures in conditional loss (structure-to-structure spread)?
- At what buffer radius do fuel measures matter most (30 m vs 500 m)?
- Do terrain effects persist after controlling for fuels and density?
- Do importance rankings differ between the held-out recent fire and the training fires?

These are hypotheses to test, not results.

Repository, reproducibility and limitations

Every stage is a scripted, cached step driven by one config, so the whole analysis, including the locked held-out predictions, can be rerun from raw downloads.

Repository layout

```
config/project.yaml          # CRS, buffers, fires, cutoff date, model grids
data/manifest.yaml           # sources, versions, download dates, checksums
src/wui/
  ingest.py                  # download + version pinning
  labels.py                  # DINS cleaning and footprint matching
  features.py                # buffers, zonal stats, distances
  folds.py                   # variograms, blocks, LOFO groups
  train.py                   # nested CV, Optuna, calibration
  evaluate.py                # metrics, bootstrap, maps
  explain.py                 # permutation, SHAP, ALE, stability
notebooks/                   # figures for the paper only
results/<run_id>/            # metrics, predictions, figures, config copy
```

Running

```
make data        # download and pin sources
make features    # build structures.parquet
make cv          # nested spatial CV for all models
make holdout     # train on pre-cutoff fires, score test fire, lock
predictions
make explain     # importance and SHAP
```

Environment: Python 3.11, geopandas, rasterio, exactextract, scikit-learn, xgboost, optuna, shap; pinned in `environment.yml`. Random seeds fixed per run; folds saved to disk.

Limitations

- DINS covers exposed structures only and inspection practice varies between fires, so the model estimates loss *given exposure*, not absolute risk.
- LANDFIRE is 30 m and lags ground conditions by years; it misses parcel-level vegetation and home-hardening.
- Event weather dominates many losses; a static model cannot capture it unless the weather variant is used, which then cannot be applied before a fire.
- Building attributes (roof, vents, defensible space) are often missing, which may inflate the apparent importance of correlated proxies such as density or year built.
- One or two test fires cannot establish general transferability; results are a case study.

Next steps — add exposure modelling (burn probability) to estimate unconditional risk; extend to other states; incorporate parcel-scale vegetation from high-resolution imagery.